Paper 25.50 - Energetic Traversal Claim Contract

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Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-25.50.md

Paper 25.50 - Energetic Traversal Claim Contract

Purpose

This contract defines the minimum fields required before a traversal claim can be promoted from example to paper evidence.

Required Fields

Each step must provide a path id, step id, source object, target object, transport id, `(L, C, R)` state, Noether residue, Shannon residue, Landauer cost, absorption capacity, weights `alpha`, `beta`, `gamma`, computed `theta`, unit policy, step closure status, and obligation label when present.

Each path must provide the ordered step list, `theta_path`, path closure status, open-step ids, and a Boolean statement of whether physical energy is claimed. For Paper 25 the physical-energy Boolean must remain false.

Promotion Rules

A row may be promoted when its `theta` is replayable from the NSL fields and its unit policy is declared. A path may be promoted when its total is replayable from its rows and all open rows are preserved.

The following promotions are rejected:

- analog `theta` to physical joules without calibration,
- `theta <= 0` to thermodynamic optimum,
- smaller `theta_path` to geodesic or least-action proof without a domain metric,
- NSL accounting to physical-law unification,
- uncalibrated absorption to measured absorption.

Linked Review

The paper may speak forward to later applied tools by exporting the row schema. It may speak backward to Paper 00 by strengthening the unit-honesty and obligation-carry rules. It may not hide open lifts behind positive language.